MILITARY SPECIFICATION

INTEGRATED CIRCUIT , DIGITAL, CMOS, TIME BASE AND SPECIAL PURPOSE FUNCTION GENERATOR, MONOLITHIC, SILICON

This Amendment forms a part of Military Specification MIL-I-48331A(AR), dated 15 November 1979, and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1: Add under standard:

Federal
FED-STD-595 - Colors

Under Drawings delete "Tests Vehicle, Ballistic Rail Gun, 9290426, Housing, Timing and Fuzing, 9298450"

3.1.1: Add the following paragraph:

"3.1.1 Atmospheric control, storage and handling. Manufacturer shall establish and document a control process in which all parts manufactured according to this specification must be sealed and stored in a dry atmosphere to preclude moisture entrapment within I.C. device which could cause subsequent device functional degardation or failure. This document shall be submitted 30 days prior to First Article Inspection to ARRADCOM."

PAGE 3

3.5: Delete paragraph in its entirety and substitute:

l 3.5 Solderability. Extetnal leads must be acid tin electroplated or solder dipped per MIL-M-38510, paragraph 3.5.6.2(a) or (b) and if acid tin electroplated, the minimum MIL-STD-883, Method 2003."

3.6: Delete the paragraph in its entirety and substitute:

"3.6 Resistance to solvent. The marking on the component **shall not become illegible when subjected to the resistance to** solvents testing of MIL-STD-883, Method 2015 except paragraph 2.1.d."

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3.6.1 Marking. Each component shall be marked with the last three digits of the part number and the revision letter, (Example: 5073), the manufacturing date code, and the vendor identification. The vendor may add additional marking provided they do not interfere with the required marking.

* 3.7 Delete the paragraph in its entirety and substitute:

"3.7 Bond strength. Wires from the circuit die to the component wire bods shall be capable of meeting the bond strength requirements of MIL-STD-883, Method 2011, Condition D. Pull all bonds on five (5) devices. There shall be no failure less than one (1) gram."

3.9: Pre-cap visual. Third line delete sentence from "except delete.....and 3.2.9"

3.10: Delete paragraph in its entirety.

PAGE 4

3.14.3: Fifth line delete "sixty degrees (+60°C)" and add sixty-five degrees (+65°C)"

PAGE 6

3.15: Fifth line delete "sixty degrees (+60°C)" and add sixty-five degrees (+65°C)"

3.17: Delete paragraph in its entirety.

3.18: Delete paragraph in its entirety and substitute:

"3.18: **Mechanical shock. The I.C., when mounted in an** electrically functionable and inertly loaded M692 mine in accordance with drawing 9298399, shall be capable of withstanding the shock associated with firing from a 155MM howitzer at zone eight (8), M119 charge, after being environmentally conditioned in accordance with 4.5.23. In addition to the mechanical shock of the howitzer firing, the I.C. shall survive cargo ejection and ground impact in random orientation on an uncontrolled surface which may include rocks and/or other hard matter. Complete testing shall be in accordance with 4.5.23."

3.19: Delete paragraph in its entirety and substitute:

"3.19 Data shall be recorded and rtained for all test required by this specification as specified in 4.5.17 and 4.5.18."

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3.21: Delete the paragraph in its entirety and add:

"3.21 External visual. I.C. shall meet the requirements of MIL-STD-883, Method 2009. There shall be no evidence of nonconformance with the detailed drawings, the absence of any required feature, damage, corrosion or contamination."

PAGE 7

3.22.a: First line after "or" add "material or"

3.22.c: Delete subparagraph in its entirety.

3.22.d: Delete subparagraph in its entirety.

PAGE 8

4.3.3: Delete paragraph in itsd entirety and add:

"4.3.3 Mechanical shock qualification testing. Four (4) rounds each with 36 inert loaded mines shall be tested in accordance with the specified requirement of 3.18. The Technical Government Agency reserves the right not to qualify the I.D. manufacturer if abnomalities are observed during testing in accordance with 4.5.23."

Add the following paragraphs:

"4.3.4 Rejection. See Mil-A-48078.

4.3.4.1 Pass/fail criteria. Pass\fail criteria for qualification of I.C. will be as follows:

(1) Electrical and mechanical laboratory tests - The components must pass all tests to the specified sampling plan for qualification as defined on the applicable component drawings and specification, otherwise the qualification sample shall be rejected.

(2) Yuma Ballistic Test - Acceptance of the Qualification Sample shall be based upon an overall reliability of 95% at a confidence level of 90% at the component level. If performance does not achieve this level, a comparison will be made against the qualified component fired at the same time.

If the measured reliability of the Qualification Sample equals or exceeds the reliability of the qualified component, the Qualification Sample will be accepted and that vendors part be accepted for use. If the measured reliability of the Qualification Sample is significantly less (10% significance level) than the qualified component, the Qualification Sample will be rejected.

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(3) A minimum of 200 mines of the 288 ballistically fired must be recovered. Of the 200 minimum recovered, 100 mines must contain the qualified ADAM I.C. and 100 must contain the Qualification Sample. All lost mines will be considered "NO TEST" devices.

(4) An I.C. Package Failure shall be defined as a device (I.C.) failing to function electrically in which no visible I.C. Package damage is observed. This criteria should be interpreted as follows:

a. A Mine Failure attributed to a cracked I.C. in which an arc face cracked has propagated through the I.C., shall constitute a NO TEST device but will be recorded and evaluated.

b. A Mine Failure attributed to a cracked I.C. in which there is no evidence of arc face cracking, shall be cosidered a rejected unit.

A Mine Failure attributed to a discrete Component (i.e. Capacitor, Thickfilm, etc.) shall be considered an acceptable unit with regard to the I.C.

d. A Mine Failure with a multitude of problems (i.e., rejectable Thickfilm Network, etc.) but also a rejectable I.C. shall be considered a failed unit with regard to the I.C.

(5) All firing data shall be retrieved and evaluated for I.e. function. No additional wight shall be plaed on any grouping of test devices (I.E., O defects allowed for high temp. units). The total defects found, out of a total sample, shall be the only criteria considered in determining overall final reliability."

PAGE 9

Delete page 9 in its entirety and substitute new page 9 attached.

PAGE 10

Delete page 10 In its entirety and substitute new page 10 attached.

PAGE 11

Delete page 1 in its entirety and substitute new page 11 attached.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Integrated Circuit, Digital, CMOS, Time Base and Special Purpose Function Generator	1 OF 3	9298507
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Resistance to Solvents	3-0-1		3.6	4.5.8
	Solderability	6-0-1		3.5	4.5.7
	Physical Dimensions	6-0-1		3.4	4.5.1
	Pre cap Visual (De-Lid)	6-0-1		3.9	4.5.6
	Bond Strength (De-Lid)	6-0-1		3.7	4.5.2
	Die Bond Shear Strength (De-Lid)	6-0-1		3.8	4.5.4
	Country of Origin		100%	3.16	Certification
	* Conditioning Screen		100%	3.12 3.13	4.5.5 4.5.9
	Electrical Parameters, Room Ambient, 25°C	77-1-2		3.14	4.5.10
	Mechanical Shock, Ballistic Test	144(a)		3.18	4.5.23
	* Integrated circuits submitted for First Article shall have been subjected to the conditioning screen and dynamic burn-in tests prior to submission and verified by the Government.				

NOTES: (a) 144 IC's are required for Yuma Proving Ground Ballistic Testing.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: High Temperature Electrical Parameters (+80°C)		SHEET 2 OF 3		DRAWING NUMBER: 9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Leakage current	52-2-3		3.15	4.5.17
	Oscillator function current	52-2-3		3.15	4.5.17
	Oscillator function output period	52-2-3		3.15	4.5.17
	"0" output voltage, terminals 13 & 15	52-2-3		3.15	4.5.17 Note 1
	Sequential operation	52-2-3		3.15	4.5.17
	LVD function voltage	52-2-3		3.15	4.5.17
	"0" output voltage, terminal 12	52-2-3		3.15	4.5.17 Note 2

NOTES: (1) Measure after applying 4.0V (VDD) and then ramping VDD to 3.0V before 1/6 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Low Temperature Electrical Parameters (-55°C)		SHEET 3 OF 3		DRAWING NUMBER 9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
(11)	Oscillator function current	52-2-3		3.15	4.5.17
	Oscillator function output period	52-2-3		3.15	4.5.17
	"1" output voltage terminals 13 & 15	52-2-3		3.15	4.5.17 Note 1
	Sequential operation	52-2-3		3.15	4.5.17 Note 2
	LVD function voltage	52-2-3		3.15	4.5.17

NOTES: (1) Measure after applying 4.0V (VDD) and then ramping VDD to (3.0V) before 176 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.

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PAGE 12

4.4.2.2: Delete page 12 in its entirety and substitute new page 12 attached.

PAGE 13

4.4.2.2: Delete page 13 in its entirety and substitute new page 13 attached.

PAGE 15

4.4.2.4: Title block - Delete “60°C” and substitute “65°C”

PAGE 17

4.4.2.6: Group C Environmental. Delete paragraph in its entirety.

PAGE 18

Add the following paragraphs:

“4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is descibed in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.”

PAGE 18

4.5.3: Lead Fatigue. Delete in its entirety.

4.5.6: Pre-Cap Visual. Change “2001” to "2010"

PAGE 19

4.5.8: Resistance to solvents. Change sentence from "...meet the requirements of..." to "...examine or test per MIL-STD-883, Method 2015 except paragraph 2.1.d."

* 4.5.9: Delete paragraph in its entirety and substitute:

“4.5.9 All I.C.'s shall be subjected to a dynamic burn in test of 48 hours at +125°C per MIL-STD-883, Method 1015. The I.C. shall be connected as shown in figure 18. When the test is completed, the I.C. shall be cooled to twenty-five degrees centigrade (+25°C) and the electrical tests delineated in Table II shall be performed.”

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4.5.17: High temperature electrical parameters. Change temperature "60°C" to "65°C"

4.5.19: Group C Environmental Test. Delete paragraph in its entirety.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Integrated Circuit, Digital, CMOS, Time Base and Special Purpose Function Generator (physical properties)	1	1	9298507
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Physical Dimensions	3-0-1		3.4	4.5.1
102	Solderability, 5 Devices, All 16 Leads	8-0-1	1/	3.5	4.5.7
103	Bond Strength, 5 Pieces, All 16 Bondwires	8-0-1		3.7	4.5.2
104	Die Bond Shear Strength, 5 Pieces	5-0-1		3.8	4.5.4
105	Resistance to Solvents	3-0-1		3.6	4.5.8

NOTES:
1/ There shall be no failure less than 1 gram.

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Quality conformance inspections

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Conditioning Screen Tests				9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pre-Cap Visual		100%	3.9	MIL-STD-883, Method 2010, Test Cond. B
102	Stabilization Bake		100%	3.12	MIL-STD-883, Method 1008, Test Cond. C
103	Temperature Cycling		100%	3.12	MIL-STD-883, Method 1010, Test Cond. C
104	Constant Acceleration		100%	3.12	MIL-STD-883, Method 2001, Test Cond. E
105	Seal, fine leak		100%	3.12	MIL-STD-883, Method 1014, Test Cond A or B
106	Seal, gross leak		100%	3.12	MIL-STD-883, Method 1014, Test Cond. C
107	Dynamic Burn-in		100%	3.13	4.5.9
108	Electrical Parameters, D.C., 25°C		100%	3.14	4.5.10 except para. 4.5.10.2, 4.5.10.3 and 4.5.10.4
109	External visual		100%	3.21	MIL-STD-883, Method 2009

NOTES:

(13)

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PAGE 23

4.5.23: Delete in its entirety and substitute:

"4.5.23 Mechanical shock. (Yuma Proving Ground Ballistic Testing). Fout (4) rounds containing 36 inert mines each shall be fired for evaluation. A control group of projectiles equalling the number of qualification rounds shall be fired at the same time and under the same conditions for one-to-one comparison purposes. All rounds fired at YPG for qualification shall be equally divided and preconditioned at hot (+145°F) and cold (-25°F) temperature. The rounds, with applicable ballistic charges, will be soaked at temperature for a minimum of 24 hours. They shall then be fired as quickly as possible after being removed from the temperature conditioning chamber. All mines (qualification and control) used for component qualification purposes shall be non-hazardous, short-time devices per 9298399 and meeting the requirements of MIL-P-48189A. These mines shall be loaded into projectiles at Louisiana Army Ammunition Plant."

PAGE 23

4.5.24: Delete in its entirety

4.5.25: Delete in its entirety

4.5.26: Delete in its entirety

4.5.27: Delete in its entirety

PAGE 24

4.5.28: Delete in its entirety

PAGE 28

Delete Table III and substitute new Table III attached.

Add attached page 28-A, Table 4, to follow page 28.

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TABLE III

ELECTRICAL CHARACTERISTICS, HIGH & LOW TEMPERATURE
+65°C and -40°C
LOT ACCEPTANCE TESTS

CHARACTERISTIC	CONDITIONS	LIMITS
LEAKAGE CURRENT	TA = +65°C, VDD = 4.0v	16 u_a max.
OSCILLATOR FUNCTION CURRENT (SHALL BE THE AVERAGE VALUE MEASURED OVER AT LEAST FIVE (5) PERIODS	TA = -40°C, VDD = 4.0v TA = +65°C, VDD = 4.0v	50 u_a max. 50 u_a max.
OSCILLATOR FUNCTION OUTPUT PERIOD	TA = +25°C, VDD = 3.5v TA = +25°C, VDD = 4.0v TA = -40°C, VDD = 3.5v TA = -40°C, VDD = 4.0v TA = +65°C, VDD = 3.5v TA = +65°C, VDD = 4.0v	F_1 F_2 +5% F_1 & F_2 (25°C) -7% See Para Req't (3.14.3)
"1" OUTPUT VOLTAGE TERMINALS 13, 15	TA = -40°C, VDD = 3.0v MEASURED AFTER APPLYING 4.0v THEN LOWER VDD to 3.0v	1.5 volts min.
"0" OUTPUT VOLTAGE TERMINALS 13, 15	TA = +65°C, VDD = 3.0v MEASURED AFTER APPLYING 4.0v, THEN LOWER VDD TO 3.0v AND BEFORE 176 PULSES HAVE BEEN APPLIED	1.5 volts max.
SEQUENTIAL OPERATION	TA = -40°C, VDD = 3.0v TA = -40°C, VDD = 4.5v TA = +65°C, VDD = 3.0v TA = +65°C, VDD = 4.5v	See Para Req't (3.14.9)
LVD FUNCTION	TA = -40°C TA = +65°C	1.5 - 3.5 volt 1.5 - 3.5 volt
"0" OUTPUT VOLTAGE TERMINAL 12	TA = +65°C, VDD = 3.0v VDD = 3.0 VOLTS OR 0.5 VOLT ABOVE LVD WHICHEVER IS GREATER	2.0 volts max.

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TABLE IV

ELECTRICAL CHARACTERISTICS, HIGH & LOW TEMPERATURE, +80°C and -55°C
LOT ACCEPTANCE TESTS

CHARACTERISTIC	CONDITIONS	LIMITS
LEAKAGE CURRENT	TA = +80°C, VDD = 4.0v	16 u_a max.
OSCILLATOR FUNCTION CURRENT (SHALL BE THE AVERAGE VALUE MEASURED OVER AT LEAST FIVE (5) PERIODS	TA = -55°C, VDD = 4.0v	50 u_a max.
	TA = +80°C, VDD = 4.0v	50 u_a max.
OSCILLATOR FUNCTION OUTPUT PERIOD	TA = +25°C, VDD = 3.5v TA = +25°C, VDD = 4.0v TA = -55°C, VDD = 3.5v TA = -55°C, VDD = 4.0v TA = +80°C, VDD = 3.5v TA = +80°C, VDD = 4.0v	F_1 F_2 +5% F_1 & F_2 (25°C) -7% See Para Req't (3.14.3)
"1" OUTPUT VOLTAGE TERMINALS 13, 15	TA = -55°C, VDD = 3.0v MEASURED AFTER APPLYING 4.0v THEN LOWER VDD TO 3.0v	1.5 volts min.
"0" OUTPUT VOLTAGE TERMINALS 13, 15	TA = +80°C, VDD - 3.0v MEASURED AFTER APPLYING 4.0v, THEN LOWER VDD TO 3.0v AND BEFORE 176 PULSES HAVE BEEN APPLIED	1.5 volts max.
SEQUENTIAL OPERATION	TA = -55°C, VDD = 3.0v TA = -55°C, VDD = 4.5v TA = +80°C, VDD = 3.0v TA = +80°C, VDD = 4.5v	See Para Req't (3.14.9)
LVD FUNCTION	TA = -55°C,	1.5 - 3.5 volt
	TA = +80°C, VDD = 4.2v	1.5 - 3.5 volt
"0" OUTPUT VOLTAGE TERMINAL 12	TA = +80°C, VDD = 3.0v VDD = 3.0 VOLTS OR 0.5 VOLT ABOVE LVD WHICHEVER	2.0 volts max.

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Delete Figure 13 in its entirety and add new Figure 13.

LVD FUNCTION VOLTAGE TERMINAL 15

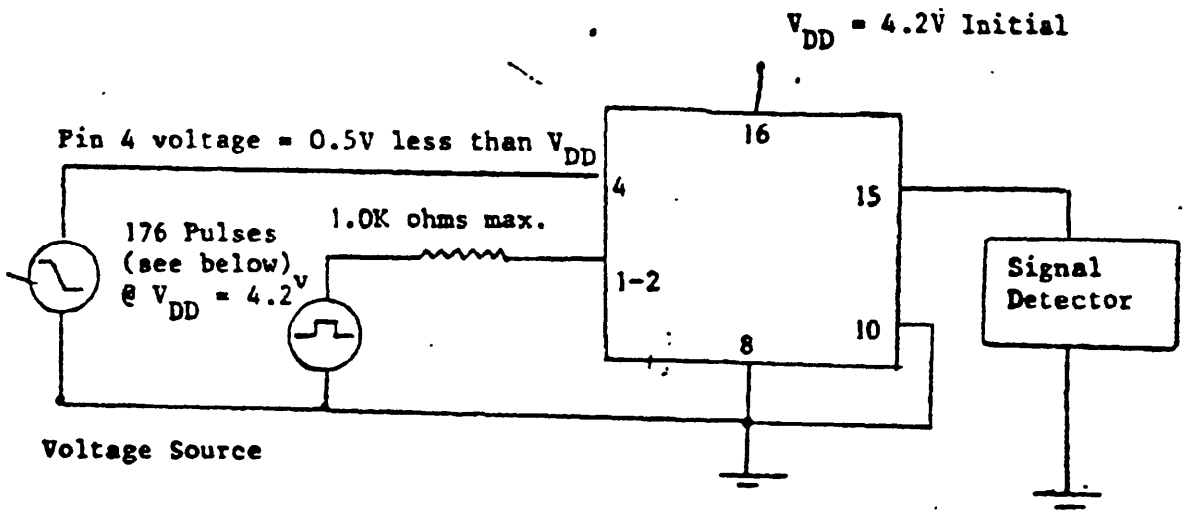


FIGURE 13

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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